



UNIVERSIDADE D
COIMBRA

Multimédia TP2

Music Information Retrieval

Membros:

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AC Report1

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1 Problem Statement

While general LLMs can answer many questions, they often lack up-to-date, specialized, or culturally specific knowledge like detailed, authentic Portuguese recipes and their variations. The challenge is building a system that can reliably and accurately answer complex, domain-specific questions about food recipes using a custom, high-quality knowledge source constructed by us using Portuguese and General food recipes datasets.

2 Project Goal

The Goal is to develop a Conversational Agent that leverages a custom structured knowledge base of food recipes built using the datasets mentioned below to provide accurate, context-aware, and conversational answers to user inquiries.

3 Data and Tools

OWL RDF for knowledge representation and reasoning. SPARQL for querying the knowledge base. Python with libraries like RDFlib for handling the RDF data. A React front-end for user interaction.

4 Methodology

Approach:

Knowledge Representation

Natural Language Processing:

5 Connection with the reviewed literature and expected outcomes

Knowledge Graph Construction/Enrichment: Knowledge Injection into LLMs: Retrieval-Augmented Generation (RAG):